SAheart: LDA/QDA Classification Analysis

Seed 43208; response chd; 9 predictors; famhist: Absent=1, Present=2.

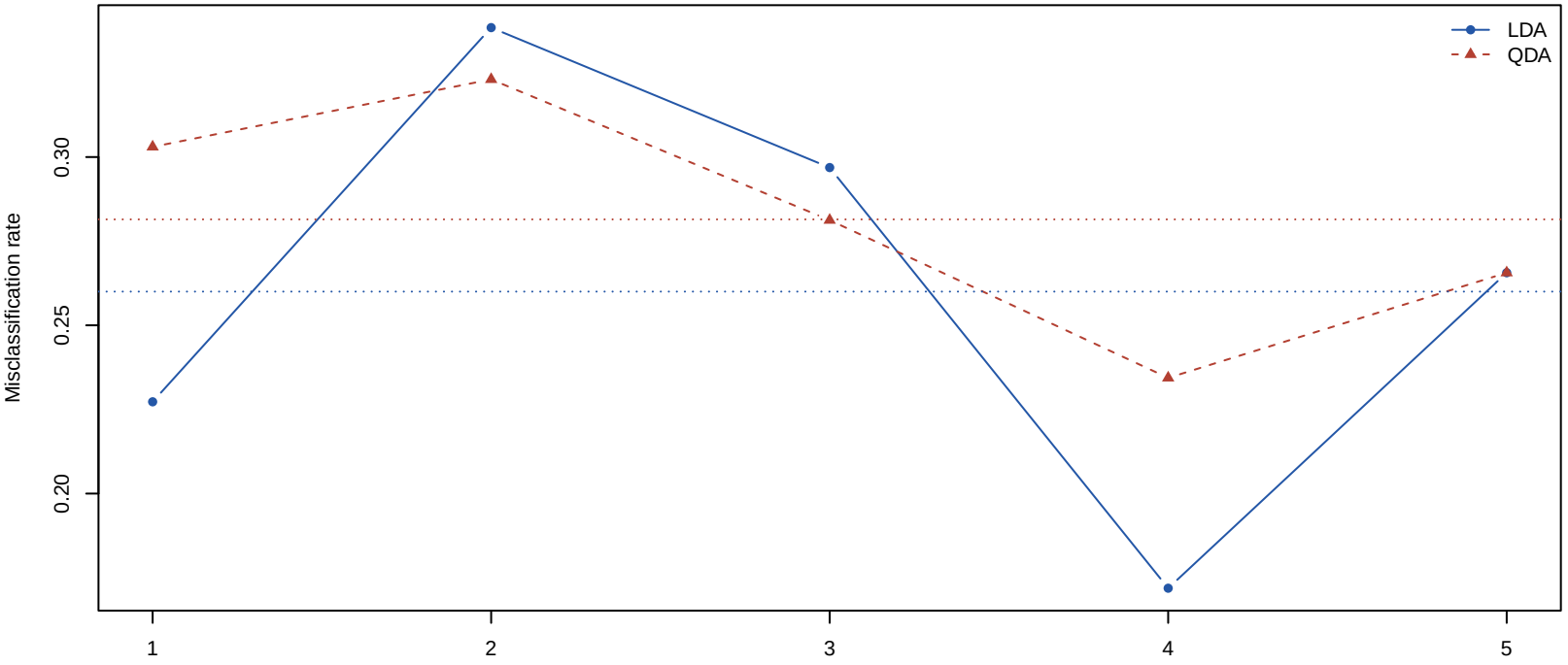
462 complete cases (302/160 by class); no rows or features excluded.

Stratified split: 323 training (211/112), 139 test (91/48).

Identical stratified 5−fold CV; scaling fitted within each training fold.

Candidate	Setting	Mean CV error
LDA	All 9 predictors	0.2600
QDA	All 9 predictors	0.2815

Both mandatory models fit successfully; all class covariance ranks were 9.



Selected by mean CV error: LDA; refitted on all 323 training cases.

One held−out evaluation: error 0.3237 (45/139); accuracy 0.6763.

Confusion matrix: observed rows / predicted columns (0, 1)

Observed 0: 68 23

Observed 1: 22 26

Class−1 sensitivity 0.542; specificity 0.747.

LDA estimates a shared covariance; QDA estimates separate class covariances.

Class imbalance and finite samples limit precision; QDA is more parameter intensive.

Full covariance rank does not guarantee stable estimates. One split and five folds

give uncertain performance estimates; results compare only these two candidates

and do not establish universal superiority.